



Caringbah High School

2018

Trial HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen only
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I 10 marks

Pages 2 – 6

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II 90 marks

Pages 7 – 15

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Question 1 - 10 (1 mark each) Answer on page provided.

1 Let $z = 4 - 3i$. What is the value of $\overline{iz}$?

A) $3 + 4i$

B) $3 - 4i$

C) $-3 + 4i$

D) $-3 - 4i$

2 What is the remainder when $x^3 + x^2 + 5x + 6$ is divided by $(x + i)$?

A) $7 - 4i$

B) $7 - 6i$

C) $5 - 4i$

D) $5 + 6i$

3 The roots of the equation $2x^3 - 3x^2 + 2x + 2 = 0$ are α, β and γ .

What is the value of $\alpha + \beta - \frac{1}{\alpha\beta}$?

A) -1

B) 1

C) $-\frac{3}{2}$

D) $\frac{3}{2}$

4 What are the coordinates of the foci of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$?

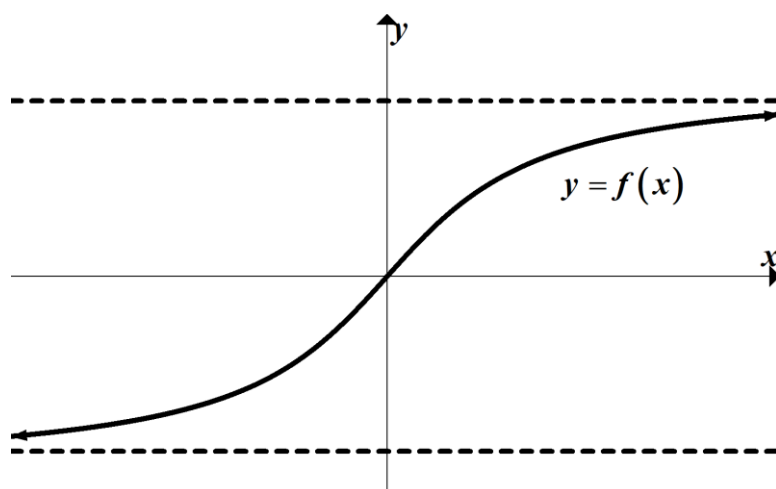
A) $\left(\pm \frac{2\sqrt{5}}{3}, 0\right)$

B) $\left(0, \pm \frac{2\sqrt{5}}{3}\right)$

C) $(\pm\sqrt{5}, 0)$

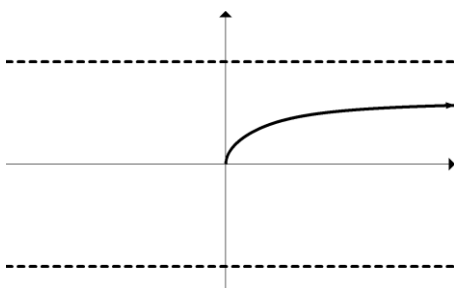
D) $(0, \pm\sqrt{5})$

- 5 The graph of $y = f(x)$ is shown below?

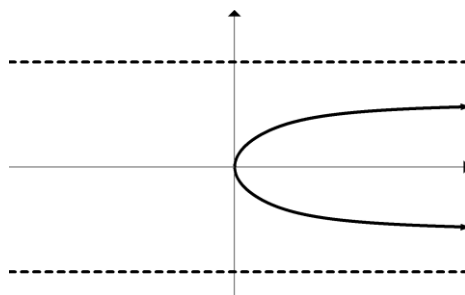


Which of the following graphs best represents $y^2 = f(x)$?

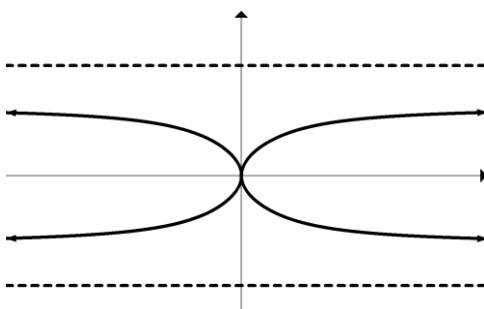
A)



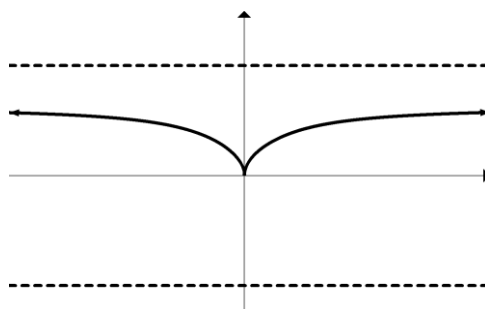
B)



C)



D)



6 Consider $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 x}{(1 + \tan x)^2} dx$.

After using an appropriate substitution, which of the following is equivalent to I ?

A) $\int_0^2 \frac{1}{u^2} du$

B) $\int_0^2 \frac{u^2}{(1+u)^3} du$

C) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u} du$

D) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{u^3} du$

7 If $e^x + e^y = 1$, which of the following is an expression for $\frac{dy}{dx}$?

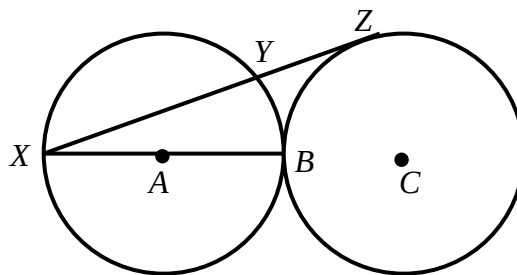
A) $-e^{x-y}$

B) $-e^{y-x}$

C) e^{x-y}

D) e^{y-x}

- 8 Two equal circles touch externally at B . XB is a diameter of the circle centred at A . XZ is the tangent from X to the circle centred at C and cuts the first circle at Y .



Which is the correct expression that relates XZ to XY ?

A) $3XZ = 4XY$

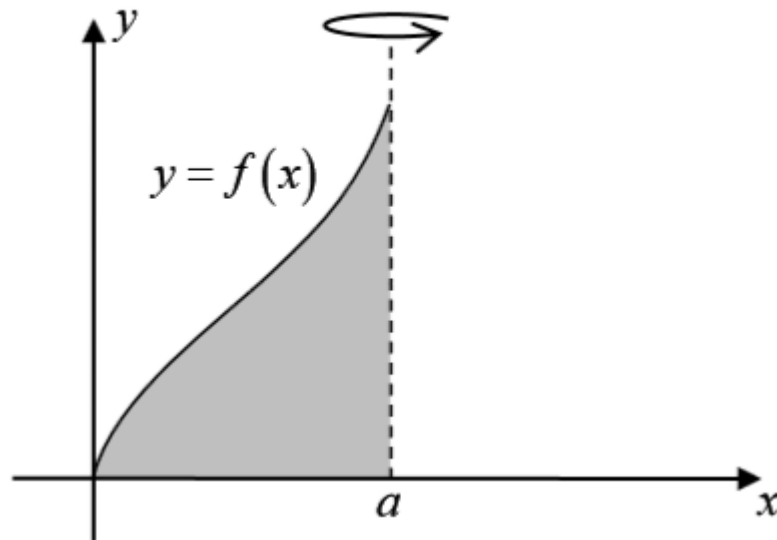
B) $XZ = 2XY$

C) $2XZ = 3XY$

D) $2XZ = 5XY$

- 9 The function $y = f(x)$ is monotonic increasing over the interval $0 \leq x \leq a$.

The region bounded by this function, the x -axis and the line $x = a$ is to be rotated about the line $x = a$ to form a solid of revolution.



Which of the following integrals represents the volume of this solid?

- A) $\pi \int_0^a [a - f(x)]^2 dx$ B) $\pi \int_0^a [a - x]^2 dx$
- C) $\pi \int_0^{f(a)} [a - f^{-1}(y)]^2 dy$ D) $\pi \int_0^{f(a)} [a - y]^2 dy$

- 10 Consider $\omega = \frac{1}{2}(-1 + i\sqrt{3})$.

Which of the following is a correct expression for $|z|$, where $z = a + b\omega$ and a and b are constants.

A) $\sqrt{(a - b)^2 + 2ab}$

B) $\sqrt{(a - b)^2 + ab}$

C) $\sqrt{(a - b)^2 - ab}$

D) $\sqrt{(a - b)^2 - 2ab}$

END OF MULTIPLE CHOICE QUESTIONS

Section II**90 marks****Attempt all questions 11–16****Allow about 2 hours and 45 minutes for this section**

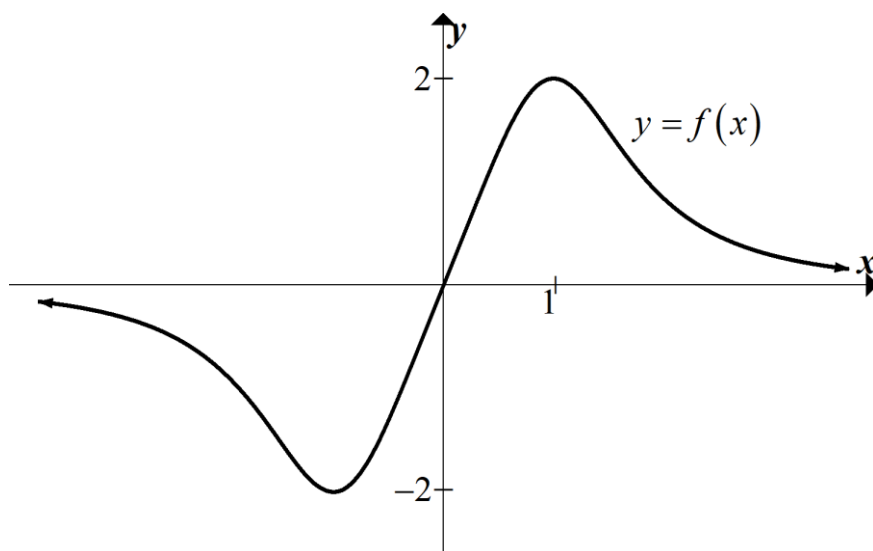
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.	Marks
a) Let $z = 1 - i$ and $u = 3 + 2i$. Find:	
(i) $\operatorname{Im}(u - z)$.	1
(ii) $z\bar{u}$.	1
b) Let the point P represent the complex number z in the Argand diagram.	1
Describe the features of the new point Q representing the complex number $2iz$.	
c) Find $\int \frac{dx}{x^2 + 6x + 13}$.	2
d) It is given that $Z = 1 - \sqrt{3}i$.	
(i) Express Z in modulus – argument form.	2
(ii) Hence, express Z^{10} in $x + iy$ form.	2
e) Find the x -coordinates of the points on the curve $2x^2 + 2xy + 3y^2 = 15$ where the tangents to the curve are vertical.	3
f) (i) Find the domain and range of $f(x) = \tan^{-1}(e^x)$.	2
(ii) Sketch $y = f(x)$ showing any intercepts and asymptotes.	1

Question 12 (15 marks) Start a NEW booklet.**Marks**

- a) The diagram below shows the graph of $y = f(x)$ which is an odd function.



On separate sketches, showing any important features, neatly draw the graphs of

- | | | |
|-------|----------------------|----------|
| (i) | $y = f(-x)$ | 1 |
| (ii) | $y = \sqrt{f(x)}$ | 2 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv) | $y = x + f(x)$ | 2 |
| (v) | $y = f'(x)$ | 2 |

Question 12 continues on page 9

Question 12 (continued)

Marks

- b) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate $\int_0^{\frac{\pi}{3}} \frac{d\theta}{1 + \sin \theta}$. 3
- c) C is a unit circle with its centre at the origin on the Argand diagram. If the point representing z_1 moves on the circle C and $z_2 = \frac{1-i}{z_1}$, find the locus of z_2 and describe it geometrically. 3

End of Question 12

Question 13 (15 marks) Start a NEW booklet.**Marks**

- a) Consider the polynomial $P(x) = x^4 - 8x^3 + 18x^2 - 27$. 3

Factorise $P(x)$, given that it has a root of multiplicity 3.

- b) Find $\int \frac{1}{x^2} \sqrt{1 + \frac{4}{x}} dx$. 3

- c) Express $\frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta + i \sin \theta}$ in the form $a + ib$. 3

- d) The roots of the cubic equation $x^3 - 4x^2 - 12 = 0$ are α, β and γ .

Find the equation with roots:

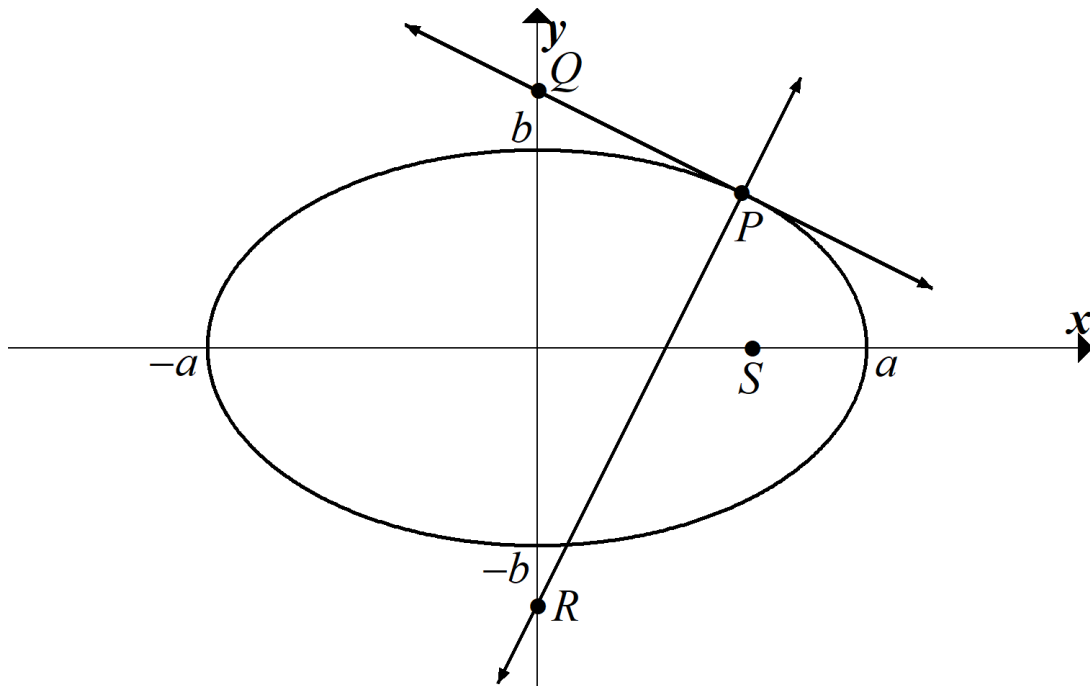
- (i) $-\alpha, -\beta, -\gamma$. 2

- (ii) $\alpha + \beta, \beta + \gamma, \gamma + \alpha$. 2

- e) Solve $a^x = e^{2x-1}$ for x , expressing the answer in terms of a . ($a > 0$). 2

Question 14 (15 marks) Start a NEW booklet.**Marks**

- a) The point $P(a\cos\theta, b\sin\theta)$ lies on the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ as shown below.



The tangent at P cuts the y -axis at Q . Also, the normal at P cuts the y -axis at R .

The normal at P is: $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$. [Do not prove this]

- (i) Show that the equation of the tangent at P is given by 2

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1.$$

- (ii) Show that the y -coordinate of Q is $\frac{b}{\sin\theta}$. 1

- (iii) Write down the coordinates of the point R . 1

- (iv) If S is the focus $(ae, 0)$, prove that P, Q, R and S are concyclic points. 3

Question 14 continues on page 12

Question 14 (continued)

Marks

- b) Consider the function $f(x) = e^{-x} - 1 + x$.

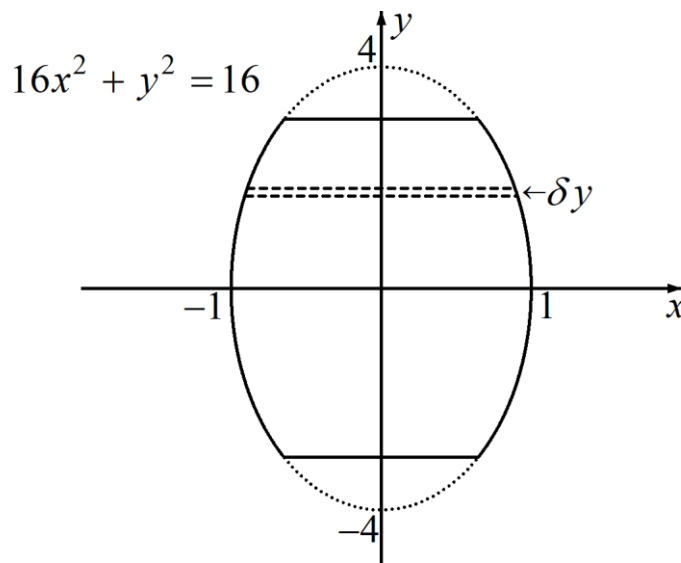
By finding expressions for $f'(x)$ and $f''(x)$ show that

3

$$e^{-x} \geq 1 - x, \text{ for all } x.$$

- c) At a children's playground there is a narrow 6 metre long tunnel with vertical walls. The base (or floor) of the tunnel is in the shape of an ellipse with equal 1 metre amounts cut from each end.

Vertical cross-sections perpendicular to the major axis of the ellipse are in the shape of a rectangle topped by a semi-circle. The base of the rectangle is twice its height and the ellipse has equation $16x^2 + y^2 = 16$.



- (i) With the aid of a diagram show that the area of a typical cross-section

2

is given by $A_s = \left(\frac{4 + \pi}{2} \right) x^2$.

- (ii) Hence, find the volume of the tunnel in exact form.

3

End of Question 14

Question 15 (15 marks) Start a NEW booklet.**Marks**

a) Find the general solution in radians to the equation $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}$. **3**

b) (i) Let x be a fixed, non-zero number satisfying $x > -1$. **3**

Use the method of mathematical induction to prove that for $n = 2, 3, 4, \dots$

$$(1 + x)^n > 1 + nx$$

(ii) Deduce that $\left(1 - \frac{1}{2n}\right)^n > \frac{1}{2}$ for $n = 2, 3, 4, \dots$ **1**

c) If $a > 0$ and $b > 0$ show that $a^3 + b^3 \geq a^2b + ab^2$. [Do not use Induction] **2**

d) $P\left(5p, \frac{5}{p}\right), p > 0$ and $Q\left(5q, \frac{5}{q}\right), q > 0$ are two points on the hyperbola $xy = 25$.

The equation of the chord PQ is given by $x + pqy = 5(p + q)$. [Do not prove this]

(i) Hence or otherwise, state the equations of the tangents at P and Q . **1**

(ii) The tangents at P and Q intersect at R . **2**

Show that the coordinates of R are given by $R\left(\frac{10pq}{p+q}, \frac{10}{p+q}\right)$.

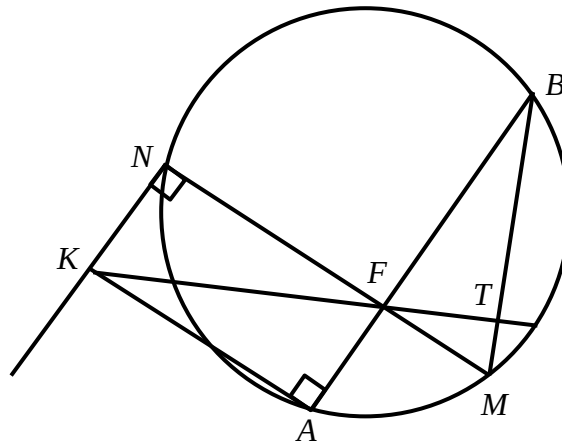
(iii) If the secant PQ passes through the point $S(15, 0)$, find the locus of R . **3**

Question 16 (15 marks) Start a NEW booklet.**Marks**

- a) The area enclosed between the curves $y = \sqrt{x}$ and $y = x^2$ is rotated about the y -axis through one complete revolution. Use the method of cylindrical shells to find the volume of the solid that is generated. **3**

- b) If $I_n = \int_0^1 x^n e^{-x} dx$, show that $I_n = -\frac{1}{e} + nI_{n-1}$. **3**

- c) As shown below, a circle has two chords AB and MN intersecting at F . Perpendiculars are drawn to these chords at A and N intersecting at K . KF produced meets MB at T .



Answer this question on the page provided

- (i) Explain why $AKNF$ is a cyclic quadrilateral. **1**
- (ii) By letting $\angle AKF = \theta$, prove that KT is perpendicular to MB . **3**

[Use and hand in the attached diagram - see page 18]

Question 16 continues on page 15

Question 16 (continued)

Marks

- d) Water flows into an aquarium at a rate proportional to the amount of water Q in the aquarium. At the same time, evaporation occurs at a rate proportional to the square of the quantity of water in the aquarium. Thus at any time ' t ', it is known that

$$\frac{dQ}{dt} = aQ - bQ^2$$

where ' a ' and ' b ' are constants. Initially $Q = Q_0$.

- i) Using partial fractions, show that an expression for Q in terms of t

4

is given by $Q = \frac{aQ_0 e^{at}}{(a - bQ_0) + bQ_0 e^{at}}.$

- ii) Show that the quantity of water tends to a limit as time increases.

1

END OF EXAM

Candidate Name/Number: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

CHS YEAR 12 EXTENSION 2 2018	TRIAL HSC SOLUTIONS
Multiple Choice Section: 1.B 2.C 3.D 4.D 5.B 6.A 7.A 8.C 9.C 10.B Question 1. $iz = i(4-3i)$ $iz = 3 + 4i$ $\therefore \bar{iz} = 3 - 4i \quad \text{-----} \boxed{B}$ Question 2. The remainder is given by $P(i) = (-i)^3 + (-i)^2 + 5(-i) + 6$ $= i - 1 - 5i + 6$ $= 5 - 4i \quad \text{-----} \boxed{C}$ Question 3. $\alpha + \beta + \gamma = \frac{3}{2}$ $\alpha \beta \gamma = -1 \rightarrow \gamma = -\frac{1}{\alpha \beta}$ $\therefore \alpha + \beta - \frac{1}{\alpha \beta} = \frac{3}{2} \quad \text{-----} \boxed{D}$ Question 4. $a=2; b=3; a^2 = b^2(1-e^2); \text{Foci } (0, \pm be)$ $\therefore e^2 = \frac{5}{9} \rightarrow e = \frac{\sqrt{5}}{3}$ $\therefore \text{Foci } (0, \pm \sqrt{5}) \quad \text{-----} \boxed{D}$ Question 5. ----- $\boxed{B}$	Question 6. let $u = 1 + \tan x \rightarrow du = \sec^2 x \, dx$ when $x = \frac{\pi}{4}, u = 2; x = -\frac{\pi}{4}, u = 0.$ $\therefore I = \int_0^2 \frac{1}{u^2} \, du \quad \text{-----} \boxed{A}$ Question 7. $e^x + e^y = 1$ and using implicit differentiation $e^x + e^y \frac{dy}{dx} = 0$ $\therefore \frac{dy}{dx} = -\frac{e^x}{e^y} = -e^{x-y} \quad \text{-----} \boxed{A}$ Question 8. $\angle XYB = 90^\circ$ and $\angle XZXC = 90^\circ.$ $\therefore \frac{XZ}{XY} = \frac{XC}{XB} = \frac{3ZC}{2ZC} = \frac{3}{2}$ $\therefore 2XZ = 3XY \quad \text{-----} \boxed{C}$ Question 9. The radius of a typical slice is $r = a - x$ $\therefore V_s = \pi r^2 \delta y$ Volume $= \pi \int_0^{f(a)} (a-x)^2 \, dy$ $= \pi \int_0^{f(a)} [a - f^{-1}(y)]^2 \, dy \quad \text{-----} \boxed{C}$

Question 10.

$$|z| = |a + b\omega|$$

$$= \left| a + b \left(\frac{1}{2}(-1 + i\sqrt{3}) \right) \right|$$

$$= \left| \left(a - \frac{b}{2} \right) + \frac{b\sqrt{3}}{2}i \right|$$

$$= \sqrt{\left(a - \frac{b}{2} \right)^2 + \left(\frac{b\sqrt{3}}{2} \right)^2}$$

$$= \sqrt{a^2 - ab + \frac{b^2}{4} + \frac{3b^2}{4}}$$

$$= \sqrt{a^2 - 2ab + b^2 + ab}$$

$$= \sqrt{(a-b)^2 + ab} \quad \text{-----} \boxed{B}$$

Question 11

a) i) $\text{Im}(u - z) = \text{Im}(2 + 3i) = 3$

ii) $z\bar{u} = (1 - i)(3 - 2i) = 1 - 5i$

b) It is rotated 90° in an anti-clockwise direction
and $|Q| = 2|P|$.

$$\begin{aligned} \text{c) } \int \frac{dx}{x^2 + 6x + 13} &= \int \frac{dx}{(x^2 + 6x + 9) + 4} \\ &= \int \frac{dx}{(x+3)^2 + 2^2} \\ &= \frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + c \end{aligned}$$

d) i) $Z = 1 - \sqrt{3}i$

$$\therefore \arg(Z) = -\frac{\pi}{3} \text{ and } |Z| = 2$$

$$\therefore Z = 2\text{cis}\left(-\frac{\pi}{3}\right)$$

d) ii) $Z^{10} = \left[2\text{cis}\left(-\frac{\pi}{3}\right) \right]^{10}$

$$= 2^{10} \text{cis}\left(-\frac{10\pi}{3}\right)$$

$$= 1024 \left[\cos\left(\frac{10\pi}{3}\right) - i \sin\left(\frac{10\pi}{3}\right) \right]$$

$$= 1024 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$= 512(-1 + \sqrt{3}i)$$

e) $2x^2 + 2xy + 3y^2 = 15$

Using implicit differentiation

$$4x + 2x \cdot y' + 2y + 6y \cdot y' = 0$$

$$2x + x \cdot y' + y + 3y \cdot y' = 0$$

$$y'(x + 3y) = -2x - y$$

$$\therefore \frac{dy}{dx} = \frac{-2x - y}{x + 3y}$$

vertical tangents will occur when $x + 3y = 0$

i.e. when $y = -\frac{x}{3}$

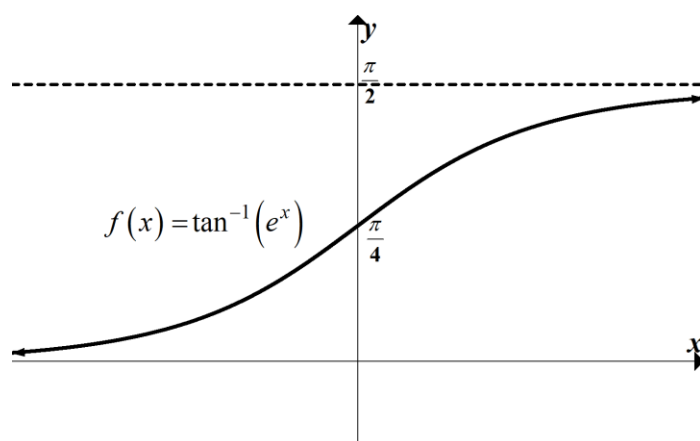
$$\therefore 2x^2 + 2x\left(-\frac{x}{3}\right) + 3\left(-\frac{x}{3}\right)^2 = 15$$

$$\therefore x^2 = 9 \rightarrow x = \pm 3$$

f) i) D: all real x .

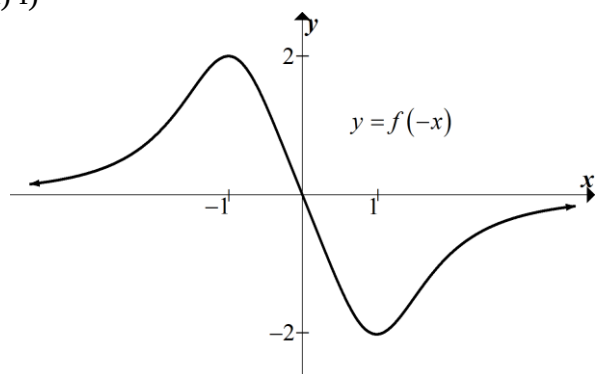
$$R: 0 < y < \frac{\pi}{2}.$$

ii)

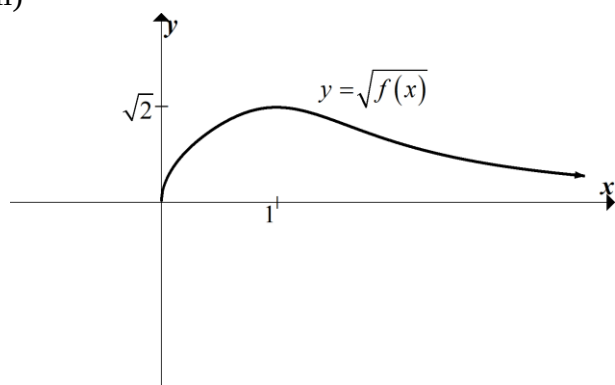


Question 12.

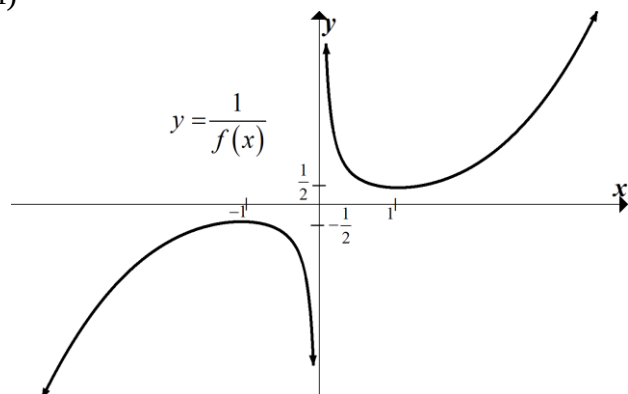
a) i)



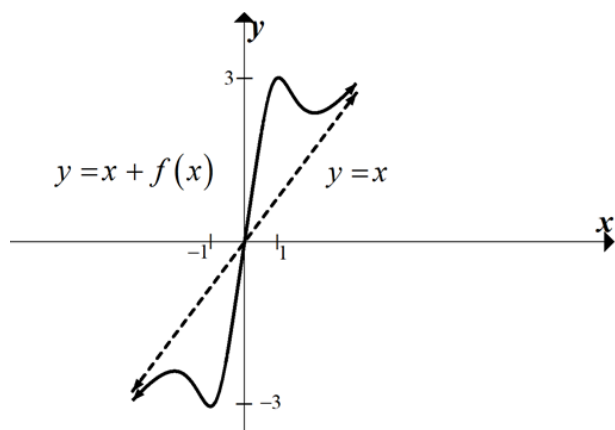
ii)



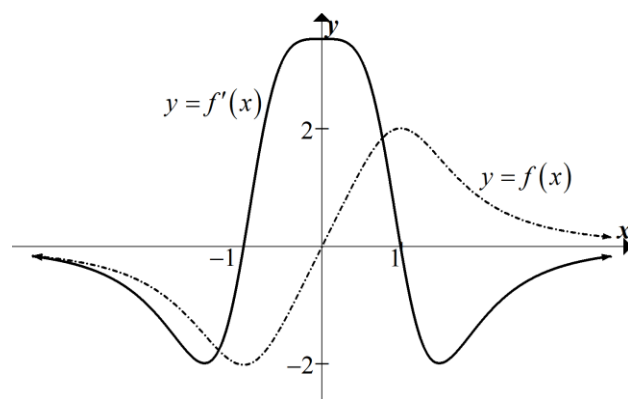
iii)



iv)



v)



$$b) \quad t = \tan \frac{\theta}{2} \rightarrow d\theta = \frac{2}{1+t^2} dt$$

$$\text{When } \theta = 0, t = 0; \theta = \frac{\pi}{3}, t = \frac{1}{\sqrt{3}}. \text{ Also } \sin \theta = \frac{2t}{1+t^2}$$

$$I = \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2t}{1+t^2} dt$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1+2t+t^2} dt = 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{(1+t)^2} dt$$

$$= -2 \left[\frac{1}{1+t} \right]_0^{\frac{1}{\sqrt{3}}} = \frac{2}{1+\sqrt{3}}$$

$$c) \quad z_2 = \frac{1-i}{z_1} \rightarrow |z_2| = \left| \frac{1-i}{z_1} \right|$$

$$\therefore |z_2| = \frac{|1-i|}{|z_1|} = \frac{\sqrt{2}}{1} \text{ as } z_1 \text{ is on the unit circle.}$$

$$\therefore |z_2| = \sqrt{2}$$

and hence the locus of z_2 is a circle centre (0,0) and radius $\sqrt{2}$.

Question 13.

a) $P(x) = x^4 - 8x^3 + 18x^2 - 27.$

$$P'(x) = 4x^3 - 24x^2 + 36x$$

$$P''(x) = 12x^2 - 48x + 36$$

For a triple root $P''(x) = 0$

$$\therefore 12x^2 - 48x + 36 = 0$$

$$\therefore 12(x-3)(x-1) = 0$$

$$\therefore x = 3 \text{ or } x = 1$$

also since $P(3) = 3^4 - 8 \times 3^3 + 18 \times 3^2 - 27 = 0$

then $x = 3$ is the root of multiplicity three.

[Note $P(1) = 1 - 8 + 18 - 27 \neq 0$]

$$\therefore P(x) = (x-3)^3(x+1)$$

b) Let $I = \int \frac{1}{x^2} \sqrt{1 + \frac{4}{x}} dx$

$$\text{Let } u = \frac{1}{x} \rightarrow du = -\frac{1}{x^2} dx$$

$$\therefore I = -\int \sqrt{1+4u} du$$

$$= -\int (1+4u)^{1/2} du$$

$$= -\frac{(1+4u)^{3/2}}{4} \times \frac{2}{3}$$

$$= -\frac{(1+4u)^{3/2}}{6}$$

$$= -\frac{1}{6} \left(1 + \frac{4}{x}\right)^{3/2} + c$$

$$c) = \frac{(1 + \cos \theta - i \sin \theta)}{(1 + \cos \theta + i \sin \theta)} \times \frac{(1 + \cos \theta - i \sin \theta)}{(1 + \cos \theta - i \sin \theta)}$$

$$= \frac{(1 + \cos \theta - i \sin \theta)^2}{(1 + \cos \theta)^2 - (i \sin \theta)^2}$$

$$= \frac{(1 + \cos \theta)^2 - 2(1 + \cos \theta)i \sin \theta + (i \sin \theta)^2}{(1 + \cos \theta)^2 - (i \sin \theta)^2}$$

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta - 2(1 + \cos \theta)i \sin \theta - \sin^2 \theta}{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}$$

$$= \frac{(\cos^2 \theta + \sin^2 \theta) + 2 \cos \theta + \cos^2 \theta - 2(1 + \cos \theta)i \sin \theta - \sin^2 \theta}{2 + 2 \cos \theta}$$

$$= \frac{2 \cos^2 \theta + 2 \cos \theta - 2(1 + \cos \theta)i \sin \theta}{2(1 + \cos \theta)}$$

$$= \frac{\cos \theta (1 + \cos \theta) - (1 + \cos \theta)i \sin \theta}{(1 + \cos \theta)}$$

$$= \cos \theta - i \sin \theta \text{ (see end for alternate solution)}$$

d) i) Let $x = -\alpha \rightarrow \alpha = -x$

$\therefore$ the required equation is given by

$$\therefore (-x)^3 - 4(-x)^2 - 12 = 0$$

$$\therefore -x^3 - 4x^2 - 12 = 0$$

$$\text{hence } x^3 + 4x^2 + 12 = 0$$

ii) The roots are of the form $\alpha + \beta + \gamma - \gamma$

$$= (\alpha + \beta + \gamma) - \gamma$$

$$= 4 - \gamma \text{ using sum of the roots.}$$

Hence let $x = 4 - \gamma \rightarrow \gamma = 4 - x$

$\therefore$ the required equation is given by

$$\therefore (4-x)^3 - 4(4-x)^2 - 12 = 0$$

$$\therefore 64 - 48x + 12x^2 - x^3 - 64 + 32x - 4x^2 - 12 = 0$$

$$\therefore x^3 - 8x^2 + 16x + 12 = 0$$

$$e) a^x = e^{2x-1}$$

$$\therefore \ln(a^x) = \ln(e^{2x-1})$$

$$\therefore x \ln a = 2x - 1$$

$$\therefore 1 = x(2 - \ln a)$$

$$\therefore x = \frac{1}{2 - \ln a}$$

Question 14.

$$a) i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ by implicit differentiation}$$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0 \rightarrow \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\therefore \text{ at } P(a \cos \theta, b \sin \theta), \frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$$

$$\therefore \text{ eq}^n \text{ of } T \text{ at } P: y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$$

$$\text{Hence } \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \text{ ----- [1]}$$

$$ii) \text{ Using [1] and putting } x = 0$$

$$\frac{y \sin \theta}{b} = 1 \rightarrow y = \frac{b}{\sin \theta}$$

$$iii) \text{ Using the equation of the normal [given] and putting } x = 0$$

$$-\frac{by}{\sin \theta} = a^2 - b^2 \rightarrow y = \frac{b^2 - a^2}{b} \sin \theta$$

$$\therefore R\left(0, \frac{b^2 - a^2}{b} \sin \theta\right)$$

$$iii) \angle QPR = 90^\circ (\angle \text{ btwn tangent and normal})$$

$$m_{QS} = \frac{0 - \frac{b}{\sin \theta}}{ae - 0} = \frac{-b}{ae \sin \theta}$$

$$m_{RS} = \frac{\frac{b^2 - a^2}{b} \sin \theta - 0}{0 - ae} = \frac{\frac{b^2 - a^2}{b} \sin \theta}{-ae}$$

$$\text{Now } b^2 = a^2(1 - e^2) \rightarrow b^2 - a^2 = -a^2 e^2$$

$$\therefore m_{RS} = \frac{-a^2 e^2 \sin \theta}{-abe} = \frac{ae \sin \theta}{b}$$

$$\therefore m_{QS} \times m_{RS} = \frac{-b}{ae \sin \theta} \times \frac{ae \sin \theta}{b} = -1$$

$$\therefore \angle QSR = 90^\circ = \angle QPR$$

Hence QPSR are concyclic points
[$\angle$'s in the same segment]

$$b) f(x) = e^{-x} - 1 + x$$

$$\therefore f'(x) = -e^{-x} + 1 \text{ and } f''(x) = e^{-x}$$

$$\text{As } f''(x) = e^{-x} > 0 \text{ for all } x$$

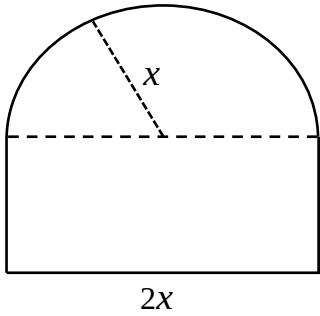
then $f(x)$ is concave up

$$\text{Also } f'(x) = 0 \rightarrow e^{-x} = 1$$

$$\therefore x = 0 \text{ and hence there is a minimum turning pt at } (0, 0)$$

$$\therefore e^{-x} - 1 + x \geq 0 \rightarrow e^{-x} \geq 1 - x.$$

c) i)



$$A = 2x \times x + \frac{\pi x^2}{2}$$

$$= 2x^2 + \frac{\pi x^2}{2}$$

$$= \left(2 + \frac{\pi}{2}\right) x^2$$

$$= \left(\frac{4 + \pi}{2}\right) x^2$$

$$\begin{aligned}
\text{cii) } V &= \left(\frac{4 + \pi}{2} \right) \int_0^3 2x^2 dy \\
&= (4 + \pi) \int_0^3 \frac{16 - y^2}{16} dy \\
&= \left(\frac{4 + \pi}{16} \right) \int_0^3 16 - y^2 dy \\
&= \left(\frac{4 + \pi}{16} \right) \left[16y - \frac{y^3}{3} \right]_0^3 \\
&= \left(\frac{4 + \pi}{16} \right) (48 - 9) \\
&= \frac{39}{16} (4 + \pi) \text{ m}^3
\end{aligned}$$

Question 15.

$$\text{a) } \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} = \frac{3}{4}.$$

$$\therefore \frac{(s + c)(s^2 - sc + c^2)}{s + c} = \frac{3}{4}$$

$$\therefore 1 - \sin \theta \cos \theta = \frac{3}{4}$$

$$\therefore \sin \theta \cos \theta = \frac{1}{4}$$

$$\therefore \sin 2\theta = \frac{1}{2}$$

$$\therefore 2\theta = n\pi + (-1)^n \left(\frac{\pi}{6} \right)$$

$$\therefore \theta = \frac{n\pi}{2} + (-1)^n \left(\frac{\pi}{12} \right)$$

$$\text{b) i) When } n=2 \text{ } LHS = (1 + x)^2 = 1 + 2x + x^2$$

$$RHS = 1 + 2x$$

$$\text{Since } x^2 > 0, 1 + 2x + x^2 > 1 + 2x$$

Hence true for $n=2$.

Assume true for $n=k, k \geq 2$

$$\text{i.e. } (1 + x)^k > 1 + kx \text{ -----[1]}$$

Prove true for $n=k+1$

$$\text{i.e. } (1 + x)^{k+1} > 1 + (k+1)x$$

$$\text{Now } (1 + x)^{k+1} = (1 + x)(1 + x)^k$$

$$> (1 + x)(1 + kx) \text{ using [1]}$$

$$\text{and since } x > -1 \rightarrow 1 + x > 0$$

$$= 1 + kx + x + kx^2$$

$$> 1 + (k+1)x, \text{ since } kx^2 > 0.$$

Hence by induction is true for all $n \geq 2$.

$$\text{ii) Let } x = -\frac{1}{2n}. \text{ If } n \geq 2, \text{ then this satisfies the}$$

condition of part (i), including $x > -1$.

$$\therefore (1 + x)^n > 1 + nx$$

$$\therefore \left(1 - \frac{1}{2n} \right)^n > 1 - \frac{n}{2n}$$

$$= \frac{1}{2}$$

$$\therefore \left(1 - \frac{1}{2n} \right)^n > \frac{1}{2} \text{ for } n = 2, 3, 4, \dots$$

c) i.e. prove that $a^3 + b^3 - a^2b - ab^2 \geq 0$

$$\begin{aligned}\text{LHS} &= (a+b)(a^2 - ab + b^2) - ab(a+b) \\ &= (a+b)(a^2 - 2ab + b^2) \\ &= (a+b)(a-b)^2 \geq 0, \text{ since } a > 0 \text{ and } b > 0. \\ \therefore a^3 + b^3 &\geq a^2b + ab^2\end{aligned}$$

d) i) By letting $p = q$ in $x + pqy = 5(p+q)$ the equations of the tangents at P and Q are $x + p^2y = 10p$ and $x + q^2y = 10q$ respectively.

ii) By solving simultaneously

$$\begin{aligned}x + p^2y &= 10p \quad \text{---[1]} \\ x + q^2y &= 10q \quad \text{---[2]}\end{aligned}$$

$$\begin{aligned}[1] - [2]: (p^2 - q^2)y &= 10(p - q) \\ (p - q)(p + q)y &= 10(p - q)\end{aligned}$$

$$\therefore y = \frac{10}{p+q} \quad (p \neq q)$$

Substituting $y = \frac{10}{p+q}$ in [1] gives

$$x + p^2\left(\frac{10}{p+q}\right) = 10p$$

$$\begin{aligned}\therefore x &= 10p - \frac{10p^2}{p+q} \\ &= \frac{10p^2 + 10pq - 10p^2}{p+q} \\ &= \frac{10pq}{p+q}\end{aligned}$$

$$\therefore R \text{ has coordinates } \left(\frac{10pq}{p+q}, \frac{10}{p+q}\right).$$

iii) Since P passes through $(15, 0)$

$$15 + 0 = 5(p+q) \rightarrow p+q = 3$$

$$\therefore R \text{ has coordinates } \left(\frac{10pq}{3}, \frac{10}{3}\right).$$

Hence the locus of R is $y = \frac{10}{3}$. [indep of p and q]

Since $p > 0$ and $q > 0$ then $pq > 0$ and so $x > 0$.

Also since the tangents intersect below the curve:

When $y = \frac{10}{3}$, $x = \frac{15}{2}$, then the locus of R is

$$y = \frac{10}{3}, \text{ for } 0 < x < \frac{15}{2}.$$

Question 16.

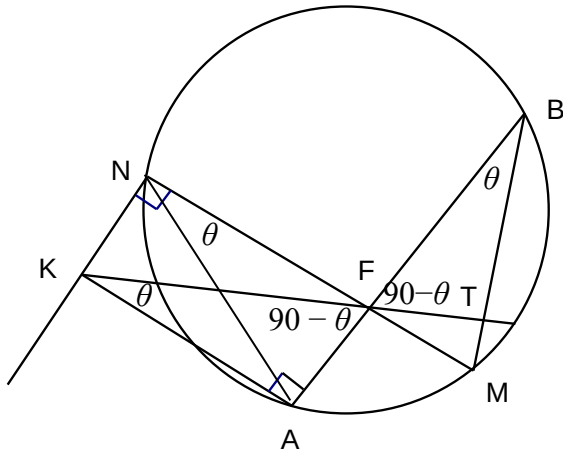
a) $V_s = 2\pi xy \delta x$

$$\begin{aligned}\therefore V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi xy \delta x \\ &= 2\pi \int_0^1 x(\sqrt{x} - x^2) dx \\ &= 2\pi \int_0^1 x^{3/2} - x^3 dx \\ &= 2\pi \left[\frac{2}{5} x^{5/2} - \frac{1}{4} x^4 \right]_0^1 \\ &= 2\pi \left(\left(\frac{2}{5} - \frac{1}{4} \right) - 0 \right) = \frac{3\pi}{10} u^3\end{aligned}$$

b) Let $u = x^n$, $v' = e^{-x}$
 $u' = nx^{n-1}$, $v = -e^{-x}$

$$\begin{aligned}\therefore I_n &= -e^{-x} x^n \Big|_0^1 + \int_0^1 nx^{n-1} e^{-x} dx \\ &= -e^{-1} - 0 + n I_{n-1} \\ &= -\frac{1}{e} + n I_{n-1}\end{aligned}$$

c) i)



Since $\angle KNF + \angle KAF = 180^\circ$

Opposite angles in cyclic quad supplementary
hence $AKNF$ is a cyclic quadrilateral.

ii) Join AN

Let $\angle AKF = \theta$

$\therefore \angle AFK = 90 - \theta$ ($\angle$ sum of $\triangle AKF$)

Also

$\angle ANF = \theta$ ($\angle$ in same segment in cyclic quad $AKNF$)

$\therefore \angle BFT = 90 - \theta$ (vertically opp $\angle AFK$)

and $\angle FBM = \theta$ ($\angle$ in same segment)

$\therefore \angle FTB = 90^\circ$ ($\angle$ sum of $\triangle FTB$)

$\therefore KT$ is perpendicular to MB .

d) i) $\frac{dQ}{dt} = aQ - bQ^2$

$$\therefore \frac{dt}{dQ} = \frac{1}{Q(a - bQ)}$$

Using partial fractions let

$$\frac{1}{Q(a - bQ)} = \frac{r}{Q} + \frac{s}{a - bQ}$$

$$\therefore 1 = r(a - bQ) + sQ$$

$$= (s - br)Q + ar$$

$$s - br = 0 \quad \text{and} \quad ar = 1 \rightarrow r = \frac{1}{a} \quad \text{and} \quad s = \frac{b}{a}$$

$$\therefore t = \int \frac{1}{a} \cdot \frac{1}{Q} + \frac{b}{a} \cdot \frac{1}{a - bQ} dQ$$

$$t = \frac{1}{a} \ln Q - \frac{1}{a} \ln(a - bQ) + c$$

$$\text{When } t = 0, Q = Q_0 \Rightarrow c = -\frac{1}{a} \ln \left(\frac{Q_0}{a - bQ_0} \right)$$

$$\therefore t = \frac{1}{a} \ln \left(\frac{Q}{a - bQ} \right) - \frac{1}{a} \ln \left(\frac{Q_0}{a - bQ_0} \right)$$

$$at = \ln \left(\frac{Q}{a - bQ} \times \frac{a - bQ_0}{Q_0} \right)$$

$$\therefore e^{at} = \frac{Q}{a - bQ} \times \frac{a - bQ_0}{Q_0}$$

$$\therefore \frac{Q_0 e^{at}}{a - bQ_0} = \frac{Q}{a - bQ}$$

$$Q = \frac{aQ_0 e^{at}}{(a - bQ_0) + bQ_0 e^{at}} \quad \text{on rearranging}$$

ii) Hence $Q = \frac{aQ_0}{bQ_0 + (a - bQ_0)e^{-at}}$

$$\text{As } t \rightarrow \infty, e^{-at} \rightarrow 0$$

$$\text{Hence } Q \rightarrow \frac{a}{b} \quad \text{which is constant.}$$

Q13c alternate solution:

Let $z = \cos \theta + i \sin \theta$ and hence $\frac{1}{z} = \cos \theta - i \sin \theta$

$$\therefore \frac{1 + \frac{1}{z}}{1 + z} = \frac{\frac{z + 1}{z}}{1 + z}$$

$$= \frac{1}{z} = \cos \theta - i \sin \theta$$